

AIMS System - Class design for all elements in the subsystem

1. Design Class: VNPaySubsystemController

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	config	Config	new Config()	Configuration for VNPay connection
2	response	Response	null	Response handler for VNPay
3	boundary	VNPayBoundary	new VNPayBoundary()	Boundary interface with VNPay

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	payOrder(amount : double, orderInfo : String)	PaymentTransaction	Initiates the payment process by creating a transaction request using the order details and amount.
2	processResponse(vnpReturnURL : String)	void	Parses and processes the payment result returned from VNPay after transaction completion.

Parameter:

- payOrder(amount: double, orderInfo: String): PaymentTransaction
 - amount:** The amount to be paid.
 - orderInfo:** Detailed information about the order.
- processResponse(vnpReturnURL: String): void
 - vnpReturnURL:** The URL containing VNPay’s response about the payment result.

Exception:

- payOrder(amount: double, orderInfo: String): PaymentTransaction
 - NotEnoughBalanceException:** This exception is thrown when the user’s bank account or linked payment method does not have sufficient funds to complete the transaction.
 - TimeoutException:** This exception is triggered when the payment request or response takes too long to process, exceeding the predefined time limit.
 - SuspiciousException:** This exception indicates that the transaction is flagged as potentially fraudulent or suspicious based on predefined security rules.
 - UnregisteredInternetBankingException:** This exception is thrown when the user tries to make an online payment using a bank account that is not registered for internet banking services.

Method:

How to use parameters/attributes:

- payOrder(amount: double, orderInfo: String): void**
 - Purpose:** This method initiates the payment process for a specific order using the provided amount and order details. It prepares a transaction request to be sent to VNPay.
 - Usage:** Called when the user confirms an order and proceeds to payment. The method sends a request to the VNPay API with the given amount and order information. Exceptions may occur if the request fails or validation errors happen.
- processResponse(vnpReturnURL: String): void**
 - Purpose:** This method extracts and validates the result of a transaction from the return URL provided by VNPay. It determines if the payment was successful or failed.
 - Usage:** Called when VNPay redirects the user back to the application. The return URL is parsed for status codes and transaction data, which are then used to finalize the payment process and update the system.

State:

- If payment is successful → transaction is created and status set to "Success".
- If payment fails → appropriate exception is raised based on error code.

2. Design Class: VNPayBoundary

Table 1. Attribute design

#	Name	Data type	Default value	Description
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1	webView	WebView	null	Web view component to display payment page
2	paymentURL	String	null	URL of the payment page

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	VNPayBoundary(paymentURL : String)	constructor	Loads and displays the VNPay payment webpage inside the application’s WebView.
2	handleURLChanged()	void	Handles URL changes in WebView to identify whether the payment process is completed.

Parameter:

- VNPayBoundary(paymentURL: String)
 - **paymentURL**: The URL of the VNPay payment interface.

Exception:

Method:

How to use parameters/attributes:

- **VNPayBoundary(paymentURL: String)**
 - **Purpose**: This method opens the VNPay payment page within the application’s WebView so that the user can complete the transaction.
 - **Usage**: Called after payOrder() to present the payment interface. It loads the paymentURL in the WebView and allows the user to perform payment actions.
- **handleURLChanged(): void**
 - **Purpose**: This method verifies whether the current WebView URL matches the expected return URL from VNPay, indicating transaction completion.
 - **Usage**: Invoked when a navigation event is detected. If the current URL matches the return URL, it triggers the processing of the payment result.

State:

3. Design Class: Config

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	vnpPayUrl	String	" https://sandbox.vnpayment.vn/paymentv2/vpcpay.html "	VNPay payment URL
2	vnpVersion	String	"2.1.0"	VNPay API version
3	vnpTmnCode	String	null	Terminal ID provided by VNPay
4	vnpSecretHash	String	null	Secret key for signature generation
5	vnpReturnUrl	String	null	Return URL after payment

Table 2. Operation design

#	Name	Return type	Description (purpose)
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1	getVnplpAddress	String	Returns the server IP address
2	getRandomNumber	String	Returns the transaction number (random generate in specific time)

Parameter:

- Getters have no parameters and return their respective values.

Exception:

Method:

- Config()
 - Purpose: Initializes configuration for VNPay integration.
 - Usage: Called when creating VNPaySubsystemController.
- Various getters
 - Purpose: Provide access to configuration values.
 - Usage: Called when building payment requests.

State:

4. Design Class: Request

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	amount	double	0.0	Payment amount
2	orderInfo	String	null	Order information
3	params	Map<String,String>	null	Request parameters

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	Request(amount : double, orderInfo : String)	constructor	Constructor for Request
2	buildQueryURL()	String	Builds query URL for VNPay request

Parameter:

- Request(amount: double, orderInfo: String)
 - amount: Payment amount.
 - orderInfo: Information about the order.
- buildQueryURL(): String
 - Return: Complete URL for VNPay payment.

Exception:

Method:

- Request(amount, orderInfo)
 - Purpose: Creates a request object with payment information.
 - Usage: Called when initiating payment.
- buildQueryURL()
 - Purpose: Constructs the complete URL for VNPay payment request.
 - Usage: Called to generate payment URL.

State:

5. Design Class: Response

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	responseCode	String	null	Response code from VNPay
2	responseParams	Map<String, String>	new HashMap<>()	Response parameters
3	responseMessage	String	null	Response message from VNPay
4	secureHash	String	null	Secure hash for verification

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	Response	constructor	Initiate the response action from
2	parseQueryString	void	Parses query parameters from URL
3	handleErrorCode	void	Handles error codes in response

Parameter:

- Response(vnpReturnURL: String)
 - vnpReturnURL: The return URL from VNPay containing response data.
- parseQueryString(vnpReturnURL: String): void
 - vnpReturnURL: The URL containing query parameters to parse.

Exception:

- PaymentException: When payment fails with an error code.

Method:

- Response(vnpReturnURL)
 - Purpose: Creates a response object from VNPay return URL.
 - Usage: Called when processing payment response.
- parseQueryString(vnpReturnURL)
 - Purpose: Extracts parameters from the query string.
 - Usage: Called when processing return URL.
- handleErrorCode()
 - Purpose: Checks response code and throws appropriate exceptions.
 - Usage: Called after parsing query string.

State:

- If response code indicates success → processing continues normally.
- If response code indicates error → appropriate exception is thrown.

This completes the detailed class design for all major components in the VNPay subsystem. Each class has been documented with its attributes, operations, parameters, exceptions, methods, and states following the same format as your PaymentTransaction example.